

WIRE COLOR CODE (1 OF 2)

Color tells you what the wire is. The spec tells you what to pull.

	4 8 0 V P O W E R 8 AWG THHN/MTW Cu (40 A circuit) taps 14 AWG · SPD leads 10 AWG, max 0.5 m
	3 8 0 Y / 2 2 0 V A U X 10 AWG feeder + HVAC branches (Rev J) 14 AWG BMS branches · CATL AWG10 adapters at JXH1
	1 2 0 / 2 4 0 V C O N T R O L 14 AWG MTW
	2 4 V D C (+) 16 AWG MTW
	2 4 V D C C O M M O N 16 AWG MTW
	R E L E A S I N G C I R C U I T S FPL 14 AWG, one circuit per cabinet (TB14) FACP > PSN-1000 > DSI disconnect > cabinet aerosol input
	C A N + M B M U P A I R S AWG20 (0.5 mm2) twisted shielded pair twist pitch 20-30 mm · max 30 m run

WIRE COLOR CODE (2 OF 2)



CRIMP JUMPERS

AWG20 inside the MX34 plugs

J3.1-17, J902 9-12 / 10-11 - not panel wiring



ETHERNET

Cat 6A shielded minimum

CATL checklist cites Cat 7 for the ETH-module links



FIRE CIRCUITS

FPL 14-18 AWG per Potter / NFPA 72

CATL fire cable FPL/THHN rated above 300 V



METERING (CT)

18 AWG twisted CT leads



PE / GROUND

8 AWG Cu min · EGC per NEC 250.122

rack ground 50 mm² Cu, M8 at 10 N·m (field, by others)



NEUTRALS

sized with their phase conductors

MACHINE - CHECKED

All 174 circuits carry exactly one of these classes - verified at every build.

INSIDE THE CATL CABLES

CONDUCTOR COLORS (PER CATL)

YELLOW CAN HIGH	GREEN CAN LOW
RED + 24 V	BLACK 0 V
RED LINE (L)	BLACK NEUTRAL (N)
WHITE ALL OTHERS	

These are the colors INSIDE the CATL-supplied harnesses -
different from the panel wire classes on pages 1-2.

CRIMPS + CAN BUS RULES

JAE MX34 CRIMPS - EXACT PARTS ONLY

SIGNAL WIRES

M34S75C4F2

every CAN / COMM / 24V wire entering an MX34 plug (0.5 mm2)

CAN SHIELD

M34S75C4F3

the shield gets its OWN dedicated contact - never the signal crimp

CRIMP RULE

20 AWG

at all connector crimps, per Carson note + JAE MX34JAHL SOP

CAN BUS RULES

- Twisted shielded pair, twist pitch 20-30 mm.
- Shield grounds at ONE point only - the panel end (SB1-3). Floats at the cabinet.
- Max 30 m from MBMU to the 120-ohm end terminal; longer needs a CANBridge.
- ETH module housings bond to PE (CATL checklist item).

AS-BUILT PINOUT (1 OF 2)

From the working Carson panels, confirmed pin-by-pin against CATL's checklist.

J3.1 <-> J3.17

Crimped jumper loop - the 'plug seated' key CATL requires.

J3.2 / J3.19

MBMU +24 V / 0 V, from panel fuse BMSn. Checklist expects ~24 V

here.

J3.6 / J3.23

NOT WIRED (marked no-connect, Rev K). Fire release is Option B -

the fire panel drives the cabinet aerosol input directly. Never

wire these AND the FACP circuits.

J5.13 / J5.14

MCAN High / Low to rack JX1_IN 9/10. Yellow / green, twisted.

J5.29 / J5.30

Pair A, to ETH module J902.6 / J902.5.

AS-BUILT PINOUT (2 OF 2)

From the working Carson panels, confirmed pin-by-pin against CATL's checklist.

J5.33 / J5.34

Pair B, to ETH J902.22 / J902.21 - with a 60-ohm resistor

crimped at the MBMU end (Carson as-built).

J902.1 / J902.13

ETH module +24 V / 0 V, panel fuse 'ETH'.

J902 9-12, 10-11

Crimped jumpers - engage the ETH-internal termination per CATL.

J902 RJ45

Ethernet to switch SW1, VLAN 30, Cat 6A shielded.

ETH housing

Bonds to PE (CATL checklist item).

THE 60 - OHM COMMISSIONING CHECK

Measure across J902 pins 21/22: expect ~60 ohms. If it reads

~30 ohms, pair B is over-terminated: remove the crimped 60R

(R1-R3). CATL ICD item - measure, don't guess.

POWER ARCHITECTURE

W H O R U N S O N W H A T

M B M U + E T H M O D U L E S

24 VDC from THIS panel: redundant PSU3+PSU4 pair via ORing module

(Rev K). They survive an aux outage AND a supply failure.

R A C K B M S + C H I L L E R S

220 V aux via JXH1: A/B chiller, C/D BMS (through the in-rack

AC/DC converter).

A U X S O U R C E

Kehua Cab-375 internal 30 kVA 480/380 TX + 3 output breakers (Rev

I).

D C F U S E L A B E L S

BMS1/BMS2/BMS3/IO/ETH per Carson - all rated on the drawing now

(KLKR-2/-3); IO is UPS-backed (Rev K).

C H I L L E R I N R U S H - C A T L ' S D O C S D I S A G R E E

Spec R08306P05L31: 12.5 A under 1 ms. EW diagram V1.2: 31 A surge

(public sheet 32 A). Design uses 31 A worst case. 20 A C-curve

breakers hold either, and meet NEC 125% on the 13.6 A continuous

load. Verify ~230 VAC at JXH1 at commissioning.

GROUNDING + FIRE

GROUNDING - ONE BOND PER SYSTEM

- 120/240 V: bond BD1 at the CPT secondary - the ONLY bond in this panel.
- 380Y/220 V aux: bonded at the Kehua TX inside the PCS cabinet.
Verify in submittal.
- CAN shields: single-point ground at SB1-3, panel end only.

FIRE / SUPPRESSION

- Release = OPTION B (Rev K decision): FACP TRIG > PSN-1000 (own batteries, 3 A/ckt) > DS1 lockable supervised service disconnect > TB14 > cabinet JX1 aerosol input. One circuit per cabinet, EOL at the far end - never combined, never wire Option A too.
- HARDWIRED PCS TRIP (Rev K): FACP relay held energized > K1 > X23 to the Kehua EPO. Alarm, panel death, or power loss all stop the inverter - no software in the path.
- Fire branch: CB9 dedicated 15 A, red handle + lock-on (NEC 760.121(B)). Abort (constant-pressure) + manual release are supervised FIELD stations at the cabinets.
- PCS northbound control: Modbus TCP over Ethernet - now the SECOND line of defense behind the hardwired trip.

REV K CHANGES + OPEN ITEMS

- 1 Hardwired fire-to-inverter trip added (K1 relay, fail-safe, UPS-backed) - the board's #1 critical.
- 2 Fire panel moved to its own dedicated red-handled breaker CB9 (NEC 760.121(B)) - critical #2.
- 3 Fire release decided: OPTION B. PSN-1000 + service disconnect + field abort/manual-release, all supervised with EOLs in the
- 4 Class A detector loop re-drawn rack-to-rack; send and return legs in separate raceways.
- 5 24 V rebuilt: PSU3+PSU4 redundant pair for battery telemetry; supply-health contacts to the EMS (first failure annunciates,
- 6 Enclosure A/C + heater are real circuits off the 380Y aux (CB10/CB11) - not the little control transformer.
- 7 Every fuse now carries its rating and class on the drawing; CPT primaries are 3 A time-delay per NEC 450.3(B).
- 8 Fire battery math corrected: 4.73 Ah required (was misprinted 3.3) - worksheet NFPA72-W1 is the single source.
- 9 Machine checks: 174/174 across four formats + reconcilers.

STILL OPEN

- RFI-001 Kehua + RFI-002 CATL/Potter answers (EPO sense, aerosol pins, chiller nameplate).
- SCCR · NFPA72-W1 AHJ sign-off · Battalion VLAN · deviations · wire numbers · PE STAMP.

FOR REVIEW ONLY - NOT FOR CONSTRUCTION

PE review required. The one-page ARCH E schematic remains the authority.